



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Vaultify

A Secure Command-Line Password Manager in C++

Team ID: DSCPP-III-2026-T305

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T305
Semester: 3rd
Section: B / ML2 / ML3
Domain: Systems Programming
/ Utility Tools

Team Members

1. Ashraya Mishra (Lead) – 2510010429
2. Udit Joshi – 2510370397
3. Yuvraj Singh – 2510010591

Mentor

Dr. Prabhdeep Singh

Interactions completed: 2

Problem Statement & Motivation

Problem Statement

Most users maintain accounts on dozens of platforms but can only reliably remember a handful of passwords – leading to reuse and weak passwords that expose accounts to credential-stuffing after any single breach.

Motivation

- ▶ Password reuse is widespread and risky
- ▶ One breach can compromise many accounts
- ▶ Existing tools are closed or paid

Target Users / Stakeholders

- ▶ Individuals managing many accounts
- ▶ Students wanting a transparent tool
- ▶ Users seeking a local alternative

Objectives

Project Objectives

1. Build a secure, functional command-line password manager in C++.
2. Implement a hash table-based vault for $O(1)$ credential storage and retrieval.
3. Apply symmetric encryption and hash-based master password verification.
4. Ensure persistence of vault data across sessions using file handling.
5. Demonstrate practical, real-world application of core DS and OOP concepts.

Key Objective: Deliver a working, cross-platform CLI password manager that securely stores and retrieves credentials without ever storing the master password in plain text.

Proposed Solution

Proposed Approach

1. Entry – credential data class
2. Vault – hash table storage layer
3. Crypto – encryption & hashing logic
4. FileHandler – persistent storage
5. CLI – menu-driven user interface

Key Features

- ▶ Add / retrieve / update / delete / list
- ▶ Master password auth via hash verification
- ▶ Symmetric encryption of stored passwords
- ▶ Persistent, file-based vault storage
- ▶ Cross-platform (Windows & Linux)

The hash table gives $O(1)$ average lookup by site; encryption and hashed authentication ensure no credential is ever stored in plain text.

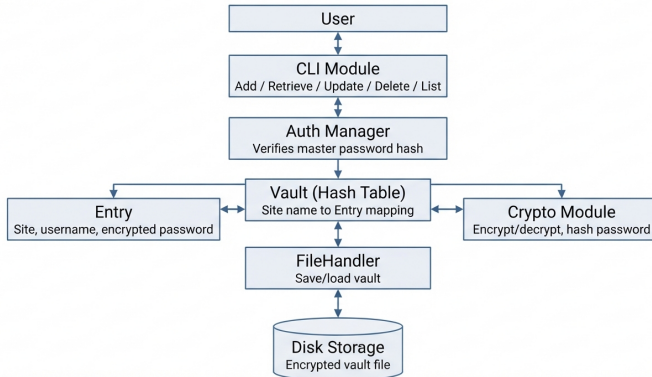
Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Hash Tables, Hash Functions & Collision Resolution, Stacks, File Structures	Vault, Recently-Accessed Stack
OOPs with C++	Classes & Encapsulation, Constructors, File Handling, Exception Handling	Entry, Crypto, File-Handler, CLI

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow

Password Manager — System Architecture



Technology Stack & Methodology

Technology Stack

- ▶ Programming: C++
- ▶ Storage: Local encrypted flat file
- ▶ Tools / Framework: Standard C++ (no external libraries)
- ▶ IDE: Visual Studio Code
- ▶ Version Control: Git

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ was chosen to directly apply DS and OOP concepts from the course; a local encrypted file avoids external dependencies, keeping the tool lightweight and fully self-contained.

Initial Progress & Team Contribution

Progress Achieved

- ▶ Problem study & existing-solution analysis completed
- ▶ Requirements and core goals finalized
- ▶ System architecture designed
- ▶ Core class structure (Entry, Vault, Crypto) scoped
- ▶ Development environment & toolchain set up

Individual Contribution

Member	Role	Contribution
Ashraya Mishra	Lead/Developer	Vault module, FileHandler, system integration
Udit Joshi	Developer	Entry class, Crypto module (encryption & hashing)
Yuvraj Singh	Developer	CLI module, output formatting, cross-platform testing

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ Fully working, cross-platform CLI manager
- ▶ Secure, encrypted local credential storage
- ▶ Clean, reusable C++ source code

References

1. Thareja, R. – *Data Structures Using C++*
2. cppreference.com – C++ & STL reference
3. GeeksforGeeks – hashing, file handling
4. OWASP Foundation – Password Storage Cheat Sheet

Thank You
Questions & Suggestions